Advertising Sales Prediction

Regression analysis of media spend and product sales

PROJECT SUMMARY

This project explores the supplied advertising dataset and compares linear regression with a random forest to predict sales from TV, radio, and newspaper budgets.

DATA AND SCOPE

200 observations; predictors are TV, radio, and newspaper advertising budgets; target is sales. A fixed 80/20 train/test split and random_state 42 make the comparison reproducible.

ANALYSIS APPROACH

Checked data types, missingness, descriptive statistics, pairwise relationships, individual spend-versus-sales patterns, and correlations. Trained a standardized linear regression baseline and a random forest regressor. Evaluated with MAE, RMSE, and R-squared, then checked held-out residuals.

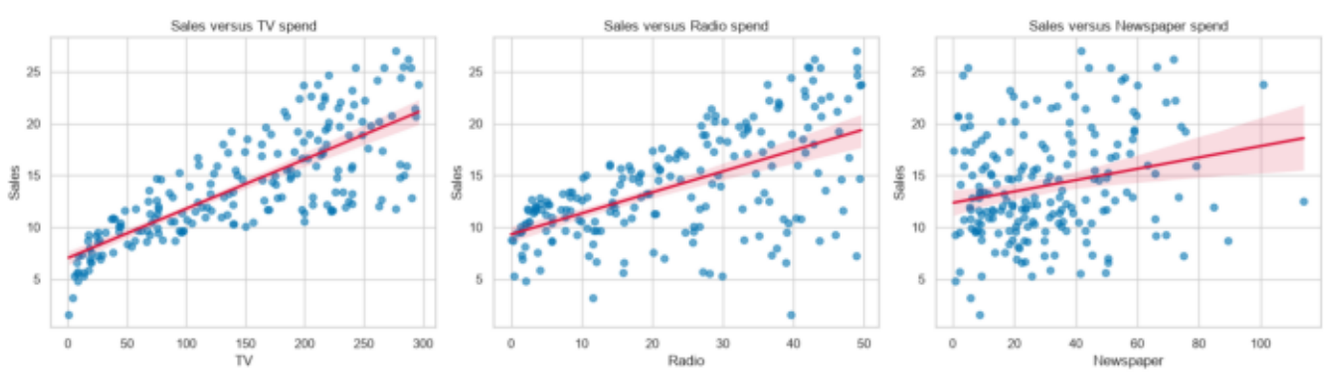
Prepared from the executed project notebook and supplied data.

RESULTS AND EVIDENCE

On the held-out split, Random Forest achieved MAE 0.662, RMSE 0.791, and R-squared 0.980; linear regression achieved MAE 1.461, RMSE 1.782, and R-squared 0.899. In the standardized linear model, TV had the largest coefficient (3.764), followed by radio (2.792); newspaper was small (0.056). These are predictive associations, not causal channel effects.



Selected notebook visualization



Selected notebook visualization

INTERPRETATION AND REPRODUCTION

Limitations and responsible use

This is a small, classic observational dataset. Results depend on the random split and may not generalize to another market, time period, or campaign. The coefficient ranking does not establish incremental causal returns.

Reproduce the analysis

Open advertising_sales_prediction.ipynb in Jupyter and run all cells. Keep Advertising Budget and Sales.csv in this folder.

The notebook includes the complete EDA, charts, modeling code, metric tables, and cell outputs.

Interpret model scores in the context of the dataset and holdout design. Observational patterns and predictive associations do not establish causal effects.